WHO target product profile for Bundibugyo virus disease vaccines

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Abbreviations and acronyms

BVD	Bundibugyo virus disease
BDBV	Bundibugyo virus
EBOV	Ebola virus
EUL	Emergency Use Listing
MARV	Marburg virus
NRA	National regulatory authorities
PQ	Prequalification
R&D	Research and development
SUDV	Sudan virus
TPP	Target product profile
WHO	World Health Organization

Background

Bundibugyo virus (BDBV) (species *Orthoebolavirus bundibugyoense*), first identified in Uganda in 2007, is a distinct species in the genus *Orthoebolavirus* within the filovirus family; it has documented human pathogenicity and mortality. Although less frequently documented than Ebola virus (EBOV) (species *Orthoebolavirus zairense*), BDBV has previously caused outbreaks. On 17 May 2026, the World Health Organization (WHO) declared that the outbreak of Bundibugyo virus disease (BVD) in the Democratic Republic of the Congo and Uganda constitutes a Public Health Emergency of International Concern (*1*). WHO underscores the need to develop BDBV-specific vaccines (*2*). Dedicated vaccine development for this species is essential both for the current BVD epidemic and as a rapidly deployable preventive tool to control future BVD outbreaks.

1. Introduction

This document serves as guidance for scientists, regulators, funding agencies, vaccine developers and manufacturers. It is intended to support stakeholders seeking WHO policy recommendations for vaccine use, as well as WHO Emergency Use Listing (EUL) or prequalification (PQ) of their products to enable access and use of BVD vaccines and meet public health needs in BVD outbreaks.

To be used in countries most affected by BVD outbreaks, vaccines will need to be evaluated and listed by WHO through EUL or PQ pathways and be included within the remit of WHO policy recommendations. An essential requirement for WHO EUL or PQ is licensure/authorization by a national regulatory authority (NRA) considered by WHO to be at maturity level 3 (stable, well-functioning and integrated) or above. WHO relies on the regulatory decisions made by WHO Listed Authorities to enable fast-track assessment of the EUL or PQ of the vaccines by WHO.

All requirements contained in WHO guidelines for policy recommendations, EUL or PQ will also apply. Considerations for WHO case-by-case assessments of BVD vaccines are listed below.

Beyond safety and an appropriate risk-benefit ratio, no single characteristic is considered more important than another. A vaccine profile that is sufficiently superior with respect to one or more critical characteristics may compensate the failure to meet a specific critical characteristic in another category. Vaccines that fail to meet multiple critical characteristics, however, are unlikely to be favourably assessed.

A generic description of WHO's vaccine prequalification and EUL processes can be found at the end of this document.

2. Bundibugyo virus disease vaccine target product profile

This document considers two scenarios for the use of a BVD vaccine, with different preferred characteristics:

a) **Reactive/outbreak use** in a BDBV outbreak to vaccinate individuals either with known BDBV exposure or at high risk of exposure, aiming to reduce/interrupt human-to human transmission to control the outbreak, to prevent the development of BDBV disease, and to reduce disease severity and mortality among infected individuals. Anticipated use will be in exposed and high-risk populations, including frontline workers and contacts of cases (implementing a ring strategy) in geographical areas experiencing, or at high risk of, an outbreak (targeted high-risk geographic areas); and in populations at high risk for importation of BVD cases from geographical areas experiencing an outbreak:

- Rapid protection: Emphasis is on rapid onset of protection that is sustained long enough to prevent disease and to interrupt transmission during an outbreak. Durability is a desirable but less critical characteristic.
- Stability/storage: In addition to requirements for storage in community health centres, long-term stability is required for stockpiling for future outbreaks of BVD. Vials for administration should be considered for ease of implementation and reduction of waste in field conditions, e.g. 2–5 doses per vial vs. 20 doses per vial.
- Risk-benefit profile: Assumes that those to be vaccinated are at high risk of exposure to BDBV, along with relatively high BVD-associated case fatality rate, currently estimated to be approximately 20%.

b) **Preventive (or prophylactic) use** to protect frontline workers (including health care workers, deploying international workers) and others at risk of BDBV occupational exposure, such as ancillary staff, lab workers, researchers working with BDBV, and those dealing with burials in the absence of ongoing outbreaks, in countries with a history of BDBV outbreaks and elsewhere.

- Durability of protection: More durable protection than for reactive/outbreak use, to prevent disease and human-to-human transmission.
- Risk-benefit profile: Assumes that some of those to be vaccinated may not be at immediate risk of exposure that is as high as the target group for reactive use.

Monovalent vs. polyvalent vaccines

In the short-term, vaccine development efforts may appropriately focus on monovalent BVD vaccines for both outbreak and preventive use. In the longer term, WHO will be seeking quadrivalent coverage for Marburg virus (MARV, species *Orthomarburgvirus marburgense*) and three species from the *Orthoebolavirus* genus—Ebola virus (EBOV), BDBV and Sudan virus (SUDV, species O. *sudanense*)—when considering filovirus vaccines for preventive use.

The term BVD is used in Table 1, but, in some cases, this refers to an anticipated polyvalent vaccine that will include a component targeting MARV and *orthoebolaviruses*.

Table 1. Target product profile (TPP) requirements for outbreak and preventive use[1]

Type of deployment	TPP for outbreak use		TPP for preventive use	
	Preferred	Critical	Preferred	Critical
Indication for use	For active immunization of near-term at-risk persons residing in the area of an ongoing outbreak for the prevention of BVD and human-to-human transmission; to be used in conjunction with other control measures to curtail or end an outbreak.		For active immunization of persons considered at risk of exposure to filoviruses for the prevention of BVD and other filovirus diseases. Broadly protective/multispecies vaccines for the members of the Filoviridae family (e.g. EBOV, SUDV, BDBV, and MARV) inclusive of BDBV are desirable.	
Target population	All age groups and special populations at high-risk of BVD in active BDBV transmission areas.	All healthy adults, excluding pregnant women, at high- risk of BDBV.	All age groups and populations and special populations at increased risk of BDBV infection and/or infection by other species of the Filoviridae family.	All healthy adults, excluding pregnant women, at increased risk of BDBV infection or infection by other species of the Filoviridae family caused by potential species of filoviruses causing future outbreaks.

[1] Critical characteristics represent the minimum acceptable profile, below which the product would not be recommended for the target use. Preferred characteristics represent the optimal profile that would maximize the public health value of the product.

Type of deployment	TPP for outbreak use		TPP for preventive use	
	Preferred	Critical	Preferred	Critical
Safety/reactogenicity	Safety and reactogenicity sufficient to provide a highly favorable risk-benefit profile in the context of observed vaccine efficacy; ideally, with only mild, transient and no serious adverse events related to vaccination.	Safety and reactogenicity consistent with expectations for a licensed/authorized vaccine for use in an individual at high risk for disease with a high mortality rate, providing an overall favorable risk-benefit profile in the context of observed vaccine efficacy.	Safety and reactogenicity at least comparable to WHO-recommended routine vaccines, providing a highly favorable risk-benefit profile, ideally with only mild or transient and no serious adverse events related to vaccination.	Safety and reactogenicity whereby vaccine benefit clearly outweighs safety risks; safety profile demonstrates primarily mild, transient health effects and only rare serious adverse events related to vaccination.
Efficacy	Greater than 80% efficacy in preventing clinical disease in healthy children, adolescents, adults, pregnant and lactating women and other special populations. Rapid onset of immunity (preferably less than 1–2 weeks).	Greater than 50% efficacy in preventing disease in healthy adults. Rapid onset of immunity (less than 21 days).	Greater than 90% efficacy in preventing disease in healthy children, adolescents and adults, pregnant and lactating women and other special populations.	Greater than 70% efficacy in preventing disease in healthy adults. If regulatory authorization is provided without clinical efficacy data, effectiveness data are to be generated during use in a future outbreak to the extent possible.

Type of deployment	TPP for outbreak use		TPP for preventive use	
	Preferred	**Critical**	**Preferred**	**Critical**
Dose regimen	Single-dose regimen highly preferred.	Single-dose regimen preferred.	Single-dose regimen preferred.	Primary series: No more than 2 doses, no more than 1 month apart. Homologous 2-dose schedules preferred over heterologous prime boost.
Durability of protection	Confers at least 2 years of protection. It may be necessary to infer long-term protection from immune kinetics.	Confers at least 6 months of protection.	Confers long-lasting protection of 5 years or more after the primary series and can be maintained by booster doses, which should be evidence informed. It is necessary to evaluate long-term protection from persistent BDBV and filovirus adaptive immunity.	Confers protection of at least 1 year after primary series and can be maintained by evidence-informed booster doses. It is necessary to evaluate long-term protection from persistent BDBV and filovirus immunity.
Route of administration	Injectable (intramuscular or subcutaneous) using standard volumes as specified in programmatic suitability for EUL/PQ or needle-free delivery. Oral or non-parenteral routes are desirable.	Injectable (intramuscular, intradermal or subcutaneous) using standard volumes as specified in programmatic suitability for EUL/PQ.	Injectable (intramuscular or subcutaneous) using standard volumes as specified in programmatic suitability for PQ or needle-free delivery. Oral or non-parenteral routes are desirable.	Injectable (intramuscular, intradermal or subcutaneous) using standard volumes as specified in programmatic suitability for EUL/PQ.

Type of deployment	TPP for outbreak use		TPP for preventive use	
	Preferred	Critical	Preferred	Critical
Species coverage	BDBV, EBOV, SUDV and MARV	BDBV	BDBV, EBOV, SUDV and MARV	BDBV
Product stability and storage	Shelf life of at least 24 months at -20 °C, and demonstration of at least 6 months' stability at 2–8 °C. The need for a preservative is determined and any issues are addressed. Vaccine vial monitor: Proof of feasibility and intent to apply a vaccine vial monitor to the vaccine. Vaccines that are not damaged by freezing temperatures (<0 °C) are preferred.	Shelf life of at least 12 months at -80 °C. Demonstrated stability for at least 8 hours at 2–8°C. The need for a preservative is determined and any issues are addressed.	Shelf life of at least 24 months at -20 °C and demonstration of at least 6 months' stability at 2-8°C. The need for a preservative is determined and any issues are addressed. Vaccine vial monitor: Proof of feasibility and intent to apply a vaccine vial monitor to the vaccine. Vaccines that are not damaged by freezing temperatures (<0 °C) are preferred.	Shelf life of at least 12 months at -20 °C. Demonstrated stability for at least 8 hours at 2–8 °C. The need for a preservative is determined and any issues are addressed.
Co-administration with other vaccines	The vaccine will be given as a stand-alone product, not co-administered with other vaccines.	The vaccine will be given as a stand-alone product, not co-administered with other vaccines.	The vaccine can be co-administered with other vaccines licensed for the same age and population groups without clinically significant impact on immunogenicity or safety of the Ebola vaccine or the co-administered vaccines.	The vaccine will be given as a stand-alone product, not co-administered with other vaccines.

Type of deployment	TPP for outbreak use		TPP for preventive use	
	Preferred	Critical	Preferred	Critical
Presentation	Vaccine is provided as a liquid product in mono-dose and multi-dose (5) presentations with a dosage volume of 0.5 mL. Multi-dose presentations should be formulated, managed and discarded in compliance with WHO's multi-dose vial policy.	Vaccine may be provided as a liquid or lyophilized product in mono-dose or multi-dose (10–20) presentations with a maximal dosage volume of 0.5 mL. Multi-dose presentations should be formulated, managed, and discarded in compliance with WHO's multi-dose vial policy. Lyophilized vaccine must be accompanied by paired separate vials of the appropriate diluent.	Vaccine is provided as a liquid product in mono-dose or multi-dose (5) presentations with a dosage volume of 0.5 mL Multi-dose presentations should be formulated, managed, and discarded in compliance with WHO's multi-dose vial policy.	Vaccine is provided as a liquid or lyophilized product in mono-dose or multi-dose (10–20) presentations with a maximal dosage volume of 0.5 mL. Multi-dose presentations should be formulated, managed, and discarded in compliance with WHO's multi-dose vial policy. Lyophilized vaccine must be accompanied by paired separate vials of the appropriate diluent.
Production capacity (the scale of each vaccine required will depend on the total number of licensed vaccines and their product attributes, cost and availability)	Process and yield scalable to produce at least 500 000 doses per year. Capacity available to manufacture and fill vaccine expeditiously as possible after scale-up. Dosage, regimen and cost of goods amenable to high volume and affordable supply.	Capacity available to manufacture at least 10 000 doses per month immediately, with evidence of potential for further scale-up.	Process and yield scalable to produce at least 500 000 doses per year. Capacity to manufacture vaccine as expeditiously as possible after scale-up. Dosage, regimen and cost of goods amenable to high volume and affordable supply.	Capacity available to manufacture vaccine as expeditiously as possible after scale-up.
Registration and EUL/prequalification	Should be recommended by WHO EUL or prequalified according to the processes outlined (*3*, *4*).			

Note: Special populations include pregnant and lactating women, paediatric, and immunocompromised populations, including persons who are HIV positive.

2. WHO Emergency Use Listing and prequalification

Emergency Use Listing

The WHO Emergency Use Listing Procedure (EUL) is a risk-based procedure for assessing and recommending unlicensed vaccines, therapeutics and in vitro diagnostics, with the ultimate aim of expediting the availability of these products to people affected by a public health emergency. This will assist United Nations procurement agencies, international organizations and Member States to determine the acceptability of using specific products, based on an essential set of quality, safety, efficacy and performance data.

The procedure is a key tool for vaccine developers and manufacturers wishing to submit their products for use during health emergencies (*4*).

Evaluation will determine whether, in light of WHO and international standards, the submitted data demonstrate a reasonable likelihood that the vaccine quality, safety and effectiveness are acceptable and that the benefits outweigh the foreseeable risks and uncertainties in the context of a Public Health Emergency of International Concern.

The submission for EUL of a vaccine dossier should follow the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use common technical document format.

Once an expression of interest is published, a vaccine manufacturer can apply to WHO and should provide the following information:

- manufacturing quality data
- nonclinical data and clinical data
- a plan to monitor quality, safety and efficacy in the field and an undertaking to submit any new data to WHO as soon as available; and
- labelling details.

Prequalification

Vaccines that are procured by United Nations agencies and for financing by other agencies, including Gavi, the Vaccine Alliance, require WHO prequalification (or EUL recommendation as indicated above). The WHO prequalification process acts as an international assurance of quality, safety, efficacy and suitability for low- and middle-income country immunization programmes. In addition, criteria were developed, such as the Programmatic Suitability for Prequalification criteria, to review vaccines submitted for prequalification (*5*).

WHO encourages vaccine developers and manufacturers to be aware of the WHO prequalification process, even at the early stages of development and to discuss the product and the regulatory requirements with the WHO prequalification staff early in the process. Developers and manufacturers need to understand WHO’s preferences for parameters that have a direct operational impact on immunization programs. Low programmatic suitability of new vaccines could result in delaying introduction and deployment. In addition, introduction of new vaccines that have higher volume, cold chain capacity or disposal demands have had a negative impact on existing operations of immunization programmes.

Registration by an NRA or the European Medicines Agency in the case of the centralized procedure for marketing authorization in Europe, will be required before any consideration of assessment for prequalification. Further, the prequalification process requires regulatory oversight by the NRA of Record, which is usually the NRA of the country where the vaccine is manufactured or the NRA of the country of finishing and distribution, and such an NRA should have been assessed as functional by WHO. Vaccine developers should check that the planned NRA of Record for the prequalification procedure is considered by WHO as maturity level 3 (stable, well-functioning and integrated) or above.

The prequalification procedure is described in detail in Annex 6 of the *WHO Expert Committee on Biological Standardization, sixty-first report* (*6*).

Beyond the minimum requirements for consideration of WHO PQ or EUL, vaccine developers should be aware of the call from immunization programmes in resource-poor settings that innovation related to programmatic suitability aspects, such as ease of administration and thermostability, will lead to great advances in these areas. Advances foreseen in the next decade include greater availability of needle-free administration for vaccine delivery in low-income countries, and thermostability so greatly improved that vaccines can be stored at ambient temperatures such that a refrigerated cold chain will no longer be needed for some vaccines. Research and collaboration between academics, vaccine and delivery device developers, together with dialogue and engagement between regulators and WHO to facilitate such advances could be transformative for immunization programmes and is strongly encouraged.

Methodology

This TPP was developed through an iterative, consultative process led by WHO, building on and adapting the existing Ebola virus TPP to reflect the specific characteristics and public health needs associated with Bundibugyo virus. The initial draft was written by the WHO research and development (R&D) Blueprint team and was subsequently revised with input from the WHO Technical Advisory Group on Candidate Vaccines Prioritization, the WHO Immunization, Vaccines and Biologicals Department and the WHO Regulation and Prequalification Department. Additional technical feedback was obtained from Gavi, the Vaccine Alliance, the Coalition for Epidemic Preparedness Innovations, and the United Nations Children's Fund. This feedback was considered by the TPP Development Group during development of the TPP in accordance with WHO procedures.

Given the Public Health Emergency of International Concern and the urgency of the outbreak, it was not possible to conduct the formal public consultation phase that would normally form part of the TPP development process. Nevertheless, broad expert input and consensus were sought throughout the review process. The WHO Research and Development Blueprint team is responsible for the final TPP.

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Declaration of interests

All external experts who reviewed and were involved in the development this document completed a WHO declaration of interest to disclose potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence concerning the subject. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects reviewed.

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